Class 11

Two player zero-sum game and both players have finite number of pure strategies.

Then $$v = \max_{\sigma_1 \in \Sigma_1} \min_{s_2 \in S_2} U(\sigma_1, s_2) = \min_{\sigma_2 \in \Sigma_2} \max_{s_1 \in S_1} U(s_1, \sigma_2)$$

For player 1, a max-min strategy is σ_1^* where

$$U(\sigma_1^*, \sigma_2) \geq v \quad \forall \sigma_2 \in \Sigma_2$$

For player 2, a max min strategy is σ_2^*, where (min-max)

$$-U(\sigma_1, \sigma_2^*) \geq -\left(\min_{\sigma_2 \in \Sigma_2} \max_{s_1 \in S_1} U(s_1, \sigma_2) = -v\right) \quad \forall \sigma_1 \in \Sigma_1$$

$$\Rightarrow U(\sigma_1, \sigma_2^*) \leq v \quad \forall \sigma_1 \in \Sigma_1$$

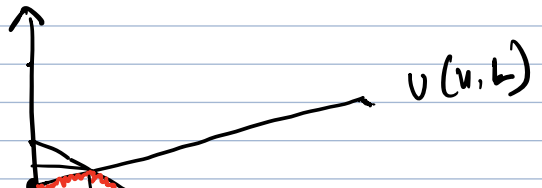
Example

P_1 \ P_2	L (y)	R ($1-y$)
T (x)	5	0
B ($1-x$)	3	4

$$v = \max_{\sigma_1 \in \Sigma_1} \min_{s_2 \in S_2} U(\sigma_1, s_2)$$

$$U(x, L) = 5x + 3(1-x) = 2x + 3$$

$$U(x, R) = 0 \cdot x + 4(1-x) = 4 - 4x$$





$2x+3 = 4-4x$

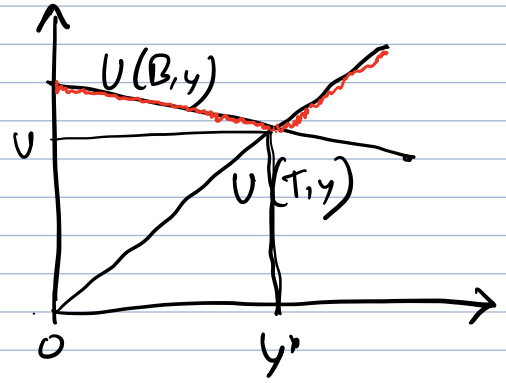
$\Rightarrow 6x = 1 \Rightarrow x = 1/6$

$x^* = \frac{1}{6}$ Unique max-min strategy for P-1

$v = 2x^*+3 = \frac{1}{3}+3 = 3\frac{1}{3}$

$U(T,y) = 5y$

$U(B,y) = 3y + 4(1-y) = 4-y$



$5y = 4-y$

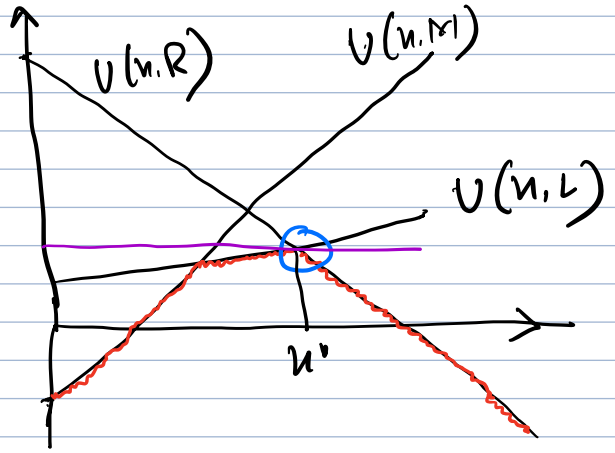
$\Rightarrow y = \frac{4}{6} = 2/3$

		L	M	R
x	T	2	5	-1
1-x	B	1	-2	5

$U(x,L) = 2x+1-x = x+1$

$U(x,M) = 5x-2(1-x) = 7x-2$

$U(x,R) = -x+5(1-x) = 5-6x$



$$U(u,R) = U(u,L) \Rightarrow 5-6u = u+1$$

$$\Rightarrow 4 = 7u$$

$$\Rightarrow \frac{4}{7} = u$$

$u^* = \frac{4}{7}$, the unique max-min strategy of P-1

$$v = U(u^*, R) = 5-6\left(\frac{4}{7}\right) = \frac{11}{7}$$

$$U(u, \sigma_2) = \sigma_2(L)\,U(u,L) + \sigma_2(M)\,U(u,M) + \sigma_2(R)\,U(u,R)$$

P-2 max-min strategy is σ_2^*

$$U(u, \sigma_2^*) \le v = \frac{11}{7} \quad \forall u$$

For a max-min strategy σ_2^*, $\sigma_2^*(M) = 0$

$$U(u,\sigma_2^*) = \sigma_2^*(L)\,U(u,L) + (1-\sigma_2^*(L))\,U(u,R)$$

$$U(0,\sigma_2^*) = \frac{11}{7}$$

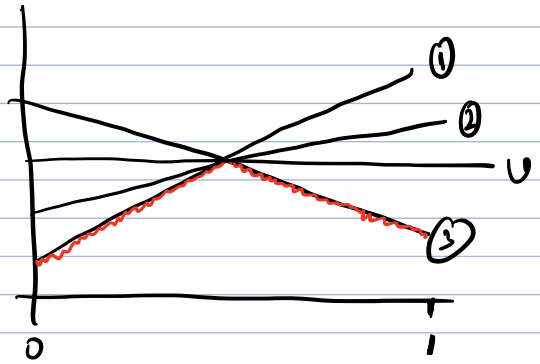
$$= \sigma_2^*(L)\,U(0,L) + (1-\sigma_2^*(L))\,U(0,R)$$

$$\frac{11}{7} \ge \sigma_2^*(L)\,1 + (1-\sigma_2^*(L))\,5$$

$$\Rightarrow \sigma_2^*(L) = \frac{6}{7}$$

For P-2, the unique max-min strategy will be

$$\left[\frac{6}{7}(L),\ 0(M),\ \frac{1}{7}(R)\right]$$



$$\alpha_1(1) + \alpha_2(2) + (1-\alpha_1-\alpha_2)(3) = v \quad \forall \alpha \in [0,1]$$

